Consistent but Blind?

Image removal does not measure image reliance in medical vision-language models

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Paper, code, data and every number

Background: is the model looking at the image?

Medical VQA benchmarks score whether an answer is right, not whether the model looked at the image. A model can answer "yes, there is a nodule" from language priors alone, and then fail exactly on the patients whose images are unusual. Paraphrase consistency does not help: a model that ignores the image is often *more* consistent, because nothing visual perturbs it.

The *consistent-but-dangerous* taxonomy (Sadanandan & Behzadan, CVPR-W 2026) therefore crosses two per-answer tests: is the answer stable under paraphrase, and does it change when the image is removed?

	changes without the image	**same** without the image
stable under paraphrase	IDEAL	DANGEROUS
flips under paraphrase	FRAGILE	WORST

A DANGEROUS answer looks trustworthy but does not depend on the image.

A taxonomy like this invites comparisons between models. We ask whether its reliance test, **removing the image**, ranks models the same way as probes that **keep an image in the input**.

Method

Three model pairs, each a medical model and the general model it was trained from (same architecture and size): **MedGemma-4B** and **MedGemma-1.5-4B** vs. **Gemma-3-4B**; **MedVLThinker-3B** vs. its base **Qwen2.5-VL-3B**.

Data. 3,365 questions on 907 images from SLAKE (CT, MRI, X-ray), VQA-RAD (radiology) and PathVQA (pathology); 1,206 yes/no test questions are the primary set.

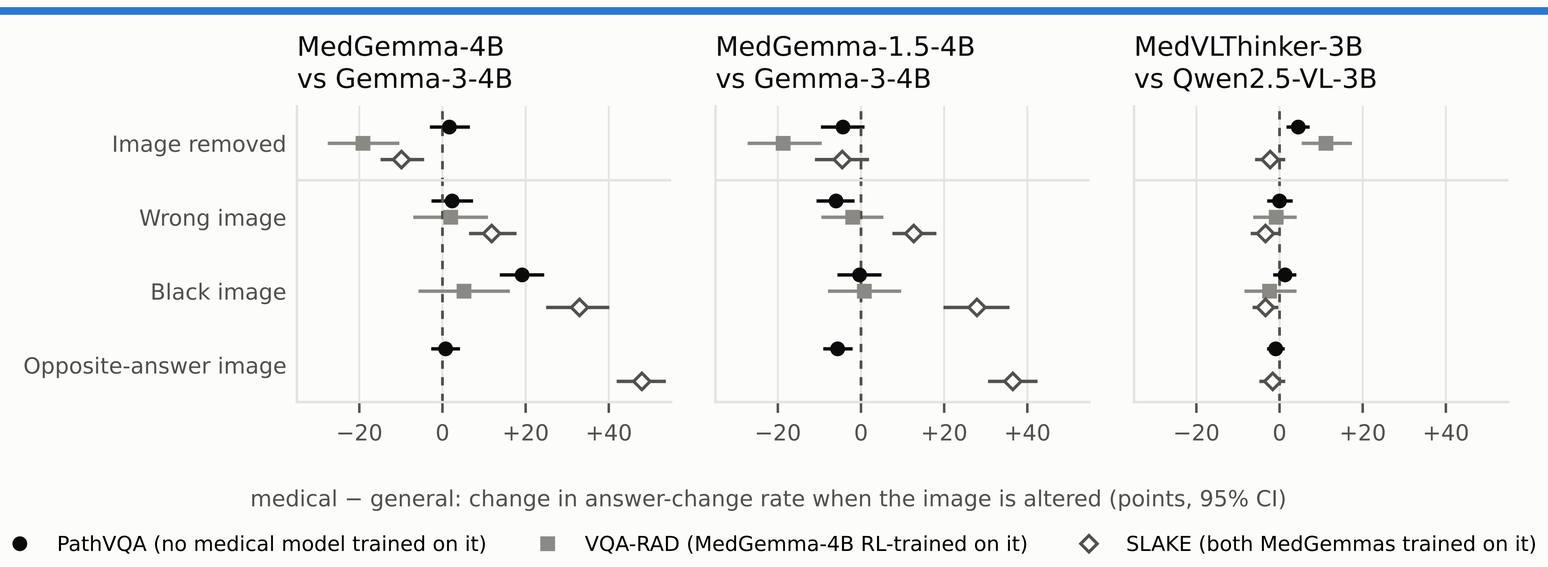
Probes. Each changes only the image or the instruction. A model that uses the image should change its answer when the image changes, so for each probe we measure the share of yes/no answers that change.

Image removed	image tokens omitted: the taxonomy's own test
Wrong image	a random image, same dataset and modality
Black image	a black image: format kept, content removed
Opposite-answer image	an image where the *same question* has the opposite answer: the most direct test
"Unknown" allowed	image removed; the model may decline

Scoring and setup. P(yes) from the yes/no token logits in one forward pass; the total yes+no probability shows when a model would rather say something else ("please provide the image"). Five models, identical prompts, bf16, greedy; about 120,000 calls on one 16 GB GPU; 95% CIs from a bootstrap over images.

Pre-registered. Endpoints frozen before the main run; every later probe and model pair registered (a dated git tag) before it ran. Failed predictions are reported.

Finding 1: image removal agrees with no probe that keeps the image



How to read it. Each point is the medical model minus its general base: the difference in how often a yes/no answer changes under one probe. Right of 0: the medical model reacts to the image more.

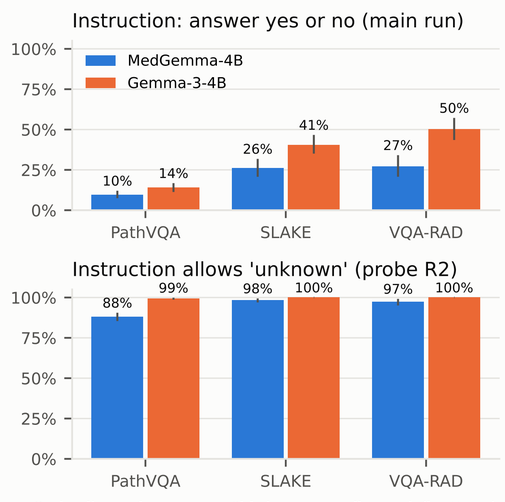
What it shows. Image removal (top row) separates the medical model from its base in five of the nine pair × dataset cells. In four, no probe that keeps an image finds any difference; in the fifth (MedGemma-4B on SLAKE), every such probe points the other way.

5 of 9
cells where image removal separates the models

0 of 5
of those confirmed by any in-format probe

Why: what image removal picks up instead

Refusal. Asked yes or no about an image it was not given, **Gemma-3** would rather ask for it on **29%** of questions, **MedGemma-4B** on 18%, **MedGemma-1.5** on 0.7%, the Qwen models never. Forced answers there are unrelated to the answers with the image ($\kappa \approx 0$), so half count as "image-reliant": refusing more looks like relying on the image more.



Right: questions declined without the image. Forced yes/no (top): Gemma-3 declines far more. "Unknown" allowed (bottom): both decline on ≥93%.

Priors that match the image. If a blind guess matches what the image shows, the answer does not change and image removal calls it blind. This explains the Qwen pair, where nobody refuses, and MedGemma-4B's extra DANGEROUS answers, which are **correct**: +14.1 points DANGEROUS-and-correct, –0.4 DANGEROUS-and-wrong.

Finding 2: the in-format advantage stays in the training data

Both MedGemmas were trained on SLAKE, where the in-format probes favoured them. We registered a test on **PathVQA**, which no medical model was trained on, before running it:

Follows the opposite-answer image medical – general, points	**SLAKE** trained on	**PathVQA** held out
MedGemma-4B – **Gemma-3**	+48.0	+0.8 [–2.7, 4.2]
MedGemma-1.5 – **Gemma-3**	+36.5	–5.6 [–9.1, –2.0]
MedVLThinker – **Qwen2.5-VL**	–1.6	–0.9 [–3.0, 1.3]

Both MedGemma predictions failed, so under the rule fixed in advance the claim that they use the image more was **withdrawn**: the SLAKE gap reflects training data. All five models do read pathology images, to a similar degree (right on both images of a swap 36–43% of the time; a model ignoring the image: 0%).

The DANGEROUS share needs correcting

- **Chance floor:** 36–42% of yes/no answers look image-independent even if unrelated to the image.
- **Correctness:** the extra DANGEROUS answers are mostly correct, from good priors.
- **No warning signal:** image reliance flags wrong answers at chance (AUROC **0.51**; confidence 0.71).

Take-aways for reliance audits

1. Don't compare models by a DANGEROUS share from image removal: it tracks refusal and priors.
2. Probe in format, ideally with an image whose correct answer is the opposite.
3. Run that probe on data outside each model's training set.
4. Report the chance floor, split by correctness, let models abstain.

Limits

Three pairs from two families: the problem exists, how common it is we cannot say. The explanations were formed after the data. The held-out test is pathology only. Paraphrases were LLM-judged (a 50-item check by the author, with AI assistance: 88%). Not a clinical claim.

Sadanandan & Behzadan, CVPR-W 2026; Zafar et al. 2026; Lotfinia et al. 2026; Sellergren et al. (MedGemma) 2025.